Lesson 5.5 Teaching Support Guide

Teach-tomorrow route

Print: current Guided Notes and Practice A for the class; Exit Tickets; optional Language Support and supported homework. Keep this guide and answer keys teacher-facing. Use the full-size model pages in the updated notes.

Phase	Planned minutes	Teaching move
Retrieve	5	Warm-Up items 1–4; note the missing prerequisite before launch.
Launch	7	Name the session target and invite a prediction from the conceptual task.
Model	10	Link a visual model to the worked example; rehearse the mathematical explanation.
Practice	18	Core: A #1, #2, #4, #6 and the error task. Choose remaining items from evidence.
Check	5	Exit Ticket; use the diagnostic page before assigning reteaching.

This is a planned 45-minute route, not a measured classroom duration. For a shorter lesson, keep the model and exit; reduce repeated practice. Form B is parallel practice or reassessment only when its items are unseen.

S1: Volume of Rectangular Prisms · I can find the volume of a rectangular prism by multiplying length, width, and height.

S2: Missing Fractional Edges · I can find a missing edge of a prism and find volumes with fractional edge lengths.

Spaced retrieval: revisit one item after 2 lessons and after approximately 1 week. Mix it with the next unit’s prerequisite check; do not announce the operation before students choose a method.

Session 1 · Volume of Rectangular Prisms

I can find the volume of a rectangular prism by multiplying length, width, and height. · 6.G.A.2

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 42 cm^3 b. 150 in^3	Build one layer first, then count equal layers along the height.
2	V = 4 × 2 × 3 = 24 ft^3	Build one layer first, then count equal layers along the height.
3	C — 120 m^3	Ask the learner to show why C — 120 m^3 is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Calls base area the volume or uses square units for volume..

Response: Build one layer first, then count equal layers along the height.

Also check: Adding the three edges instead of multiplying them.

Also check: Multiplying only two edges (finding base area) and calling it the volume.

If calculation alone is wrong, use Session 1 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 1 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A 5-by-2-by-3 m prism: find base area, number of 1 m layers, and volume.

Teacher solution: Base area 10 m²; three layers; volume 30 m³.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: One layer has 4 × 3 = 12 cubes, and two layers make 24 cubes, or 24 cm³. The base area is 12 cm²; stacking through height produces volume.

Misconception response anchor: Calls base area the volume or uses square units for volume.. Use build one layer first, then count equal layers along the height.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The relevant faces or regions are ___ because ___. My labeled dimensions show ___, so the measure is ___.

Session 2 · Missing Fractional Edges

I can find a missing edge of a prism and find volumes with fractional edge lengths. · 6.G.A.2

Observe and respond

Exit item	Expected evidence	Next prompt/model
1	a. 15 cm³ b. 84 ÷ 12 = 7 in	Name the edge length and volume of one packing cube before multiplying its count.
2	10 × 6 = 60; 180 ÷ 60 = 3 cm	Name the edge length and volume of one packing cube before multiplying its count.
3	A — 6 ft³	Ask the learner to show why A — 6 ft³ is correct and reject a specific distractor using a model.

Distinguish the error

Conceptual signal: Counts small cubes as full cubic centimeters..

Response: Name the edge length and volume of one packing cube before multiplying its count.

Also check: Dividing the volume by only one known edge.

Also check: Treating 1 1/2 as 1 (or as 1.2) when multiplying.

If calculation alone is wrong, use Session 2 Prerequisite Practice A for the matching skill, then return to the model. If the interpretation is wrong, label the quantities before reteaching the operation. Independent re-entry requires a correct setup and a connected explanation.

Session 2 · Fresh check and scoring

Reserve this check until after revision; it is distinct from the exit ticket. Do not show this teacher page to learners beforehand.

A 1-by-1-by-1 cm cube is packed with 1/2 cm edge cubes. Find count and total volume.

Teacher solution: Eight small cubes, each 1/8 cm^3; total 1 cm^3.

Score the evidence

Score	Evidence	Next action
3	Valid representation, correct result/units, connected explanation.	Continue; fade a support next time.
2	Correct setup and meaning with a calculation/notation slip, or an incomplete explanation.	Brief targeted feedback, then another unseen check.
1	Some relevant facts but a model or interpretation error.	Use the named model and the corresponding prerequisite page.
0	No interpretable evidence yet.	Oral/diagram response to separate access needs from mathematics.

Complete response anchor: 2 × 2 × 1 = 4 small cubes. Each has volume $(1/2)^3 = 1/8$ cm^3; total 4/8 = 1/2 cm^3, agreeing with the edge product.

Misconception response anchor: Counts small cubes as full cubic centimeters.. Use name the edge length and volume of one packing cube before multiplying its count.

Language access: point/draw and label → oral rehearsal → relationship sentence → independent explanation. Offer the support needed for this task and fade it based on evidence. Suggested frame: The relevant faces or regions are ___ because ___. My labeled dimensions show ___, so the measure is ___.